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Platinum nanoparticles deposited nitrogen-doped carbon nanofiber derived from bacterial cellulose for hydrogen evolution reaction

Ya Zhang [a,*], Jing Tan [a], Fangfang Wen [a], Zhifeng Zhou [a], Ming Zhu [a], Shixue Yin [a], Honggui Wang [a,b,c,**]

[a] School of Environmental Science and Engineering, Yangzhou University, Yangzhou, Jiangsu, 225127, PR China

[b] Key Laboratory of Prevention and Control of Biological Hazard Factors (Animal Origin) for Agrifood Safety and Quality, Ministry of Agriculture of China, Yangzhou University (26116120), Yangzhou, Jiangsu, 225009, PR China

[c] Jiangsu Key Laboratory of Zoonosis, Yangzhou University, Yangzhou, Jiangsu, 225009, PR China

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ABSTRACT

The nitrogen-doped carbon nanofiber derived from low and high proportion polyaniline doped bacterial cellulose (BC) was obtained via polymerization followed by pyrolysis. The resulting products were named LN-BC and HN-BC accordingly. Platinum nanoparticles modified LN-BC and HN-BC was then prepared (Pt@LN-BC and Pt@HN-BC) via electrochemical deposition. The morphologies of LN-BC and HN-BC indicated that the BC lost its nanowire structure after polyaniline modification and pyrolysis under nitrogen atmosphere. Platinum nanoparticles with diameters ranging from 3 to 5 nm can be well dispersed in the HN-BC support. The HER performance of Pt@LN-BC and Pt@HN-BC was fully investigated. Electrochemical results showed that the Pt-based catalysts had better HER activity than the Pt free catalysts in acid, indicating the HER activity was mainly from Pt. Besides, Pt@HN-BC had better HER activity than Pt@LN-BC in acid, suggesting N-doping rate was an important factor in enhancing HER activity. And the 10Pt@HN-BC (deposition for 10 s) with 4.38 wt% Pt loading was the best HER catalyst among the Pt@HN-BC. The onset potential (@ -1 mA cm^{-2}) and overpotential (@ -10 mA cm^{-2}) of the 10Pt@HN-BC in 0.5 M H_2SO_4 is -18 and -47 mV, respectively. The corresponding Tafel slope was -35 mV dec^{-1}, which is quite comparable to that of Pt/C (10 wt%). The electrochemical double layer capacitance (C_{dl}) and turnover frequency (TOF) were estimated and presented in the work. Long-term stability test confirmed that the 10Pt@HN-BC had excellent stability, which was important for practical application.

Introduction

The ever-increasing dependence on fossil fuels results in lots of undesirable situations, such as climate change, water, soil and air pollution [1–3], which have become public concerns and attracted significant attention. As one of the renewable and clean energy sources, hydrogen has been endowed with solemn mission to be the prospective counterplans to

* Corresponding author.
** Corresponding author. School of Environmental Science and Engineering, Yangzhou University, Yangzhou, Jiangsu, 225127, PR China.
E-mail addresses: zhangya@yzu.edu.cn (Y. Zhang), wanghg@yzu.edu.cn (H. Wang).

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traditional energy. Therefore, it is of great urgency to find the ways to harvest hydrogen. Among various methods, electrochemical induced hydrogen evolution reaction (HER) has been confirmed to be one of the most efficient approaches [4–7].

HER activity is mainly dependent on the HER catalysts. Therefore, researchers have paid lots of attention to prepare HER catalysts. The Pt-based or Pt-like noble metals [8–11], molybdenum-based (MoO_2, MoS_2, $MoSe_2$, MoP, Mo_2C etc.) [12–18], cobalt-based (Co_3O_4, CoS_2, CoP, $Co(OH)_2$ etc.) [19–22] and nickel-based materials ($NiSe_2$, NiFeP, NiS_2 etc.) [5,23,24] are the popular candidates for HER catalysts. Among all the HER catalysts, Pt-based materials are regarded as the highly active and effective catalysts. However, the scarcity results in the high price of Pt-based materials and finally hinders their wide applications. The promising strategies are further improving the Pt utilization efficiency and producing low Pt loading catalysts. From this view, an effective approach is to disperse Pt nanoparticles on supports with excellent electrolyte wettability, electrochemical stability and conductivity.

Nanocarbon materials, including carbon nanotubes [11,12], graphene oxide [25–28], hollow porous carbons [8] and carbon fiber cloth [29,30], have been explored as promising support to stabilize Pt nanoparticles, owing to their large surface area, excellent electrochemical stability and conductivity. For example, Chen et al. have used nitrogen-doped hollow porous carbon polyhedrons as support of HER electrocatalyst [8]. Jiang and coworkers have reported a Pt/nitrogen-doped graphene nanocomposite for the hydrogen evolution reaction [10]. As a member of nanocarbon-based family, bacterial cellulose (BC) with carbon nanotube-like structure has receiving more and more attention because it is a low-cost and eco-friendly biomass that can be easily obtained via microbial fermentation process. It has been confirmed that BC can be a good substrate or support to fabricate hierarchical nanostructures in our previous work [31]. However, the poor conductivity of BC hinders its further use as support of HER catalyst. The effective way to solve this problem is heteroatom doping. Various nitrogen-doped carbon nanofibers have been reported as the support of HER electrocatalyst with enhanced catalytic activity [18,25].

In this work, we introduced a nitrogen-doped carbon nanofibers support derived from BC using polyaniline as nitrogen source (abbreviated as LN-BC and HN-BC). The Pt modified LN-BC and HN-BC were then obtained simply via electrochemical deposition in a 0.5 M H_2SO_4 with 2 mM H_2PtCl_6 for various times. The HER activity of the obtained catalysts were fully investigated. The kinetics process and crucial parameters were estimated and presented.

Experimental section

The bacterial cellulose (BC) was bought from Hongqi Science and Technology Ltd (Guilin), moisture content was 96%. Hydrochloric acid (HCl), aniline (C_6H_7N, ANI), ammonium persulfate ($(NH_4)_2S_2O_8$, APS) and other chemicals were purchased from Sinopharm Chemical Reagent Co., Ltd unless otherwise stated. All the reagents were used as received.

Schematic of the preparation and application of 10Pt@HN-BC was summarized in Fig. 1. Before electrodeposition of Pt nanoparticles, nitrogen-doped carbon nanofiber derived from bacterial cellulose was obtained firstly. Generally, 10 g fresh and pre-cleaned BC was placed in an average of two beakers labeled A and B. Secondly, 75 mL HCl solution (1 M) poured into each beaker to immerse the BC. Then, 28 and 280 μL ANI were respectively added to A and B, stirring for 24 h to ensure ANI was fully absorbed onto BC. Next, another 75 mL HCl solution (1 M) containing different amounts (0.342 and 3.42 g) of APS (served as an oxidant for polymerization of ANI to PANI) were

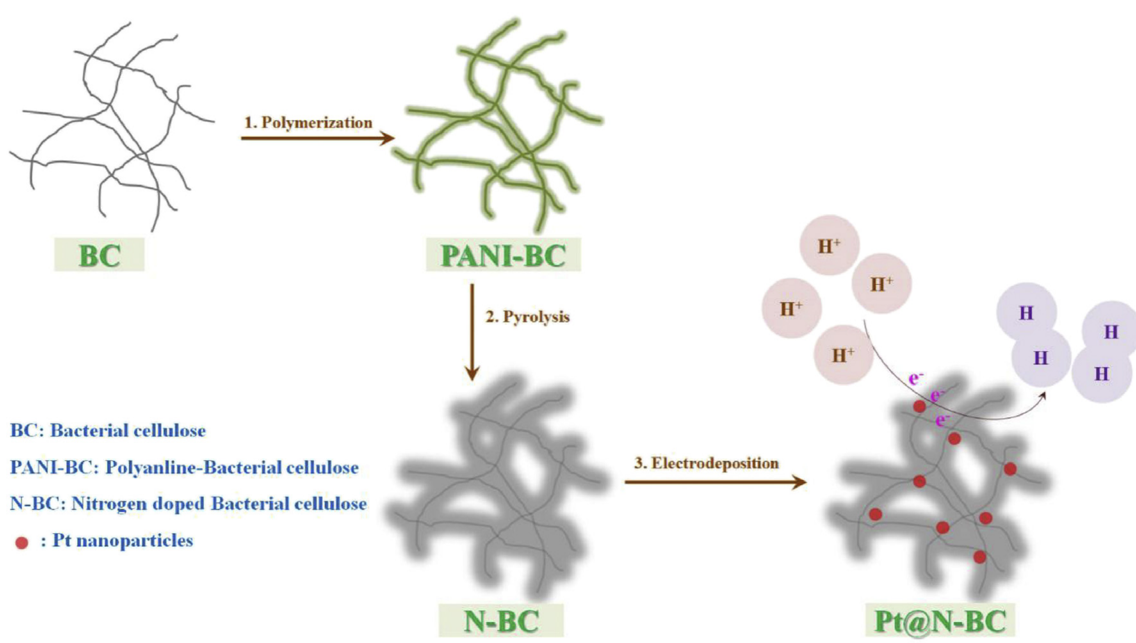

Fig. 1 – Schematic of the preparation and application of 10Pt@HN-BC.

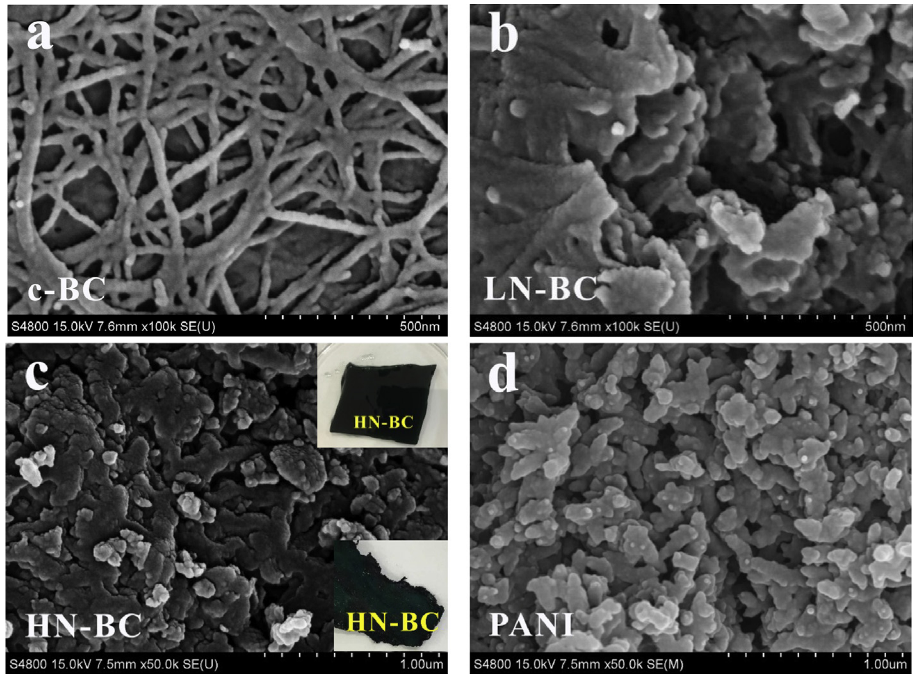

Fig. 2 – SEM images. (a) c-BC; (b) LN-BC; (c) HN-BC, insets: as-prepared and freeze-dried HN-BC; (d) PANI.

respectively added to A and B drop by drop, and stirring for another 5 h. Then, the BC with different concentrations of polyaniline were washed several times with DI water and subjected to freeze drying. Finally, the products were calcinated at 800 °C for 2 h under nitrogen atmosphere with a heating rate of 5 °C min^{-1}. After grinding and washing, the obtained powder was named as LN-BC and HN-BC, respectively. Polyaniline (PANI) was synthesized via the same steps without adding BC,

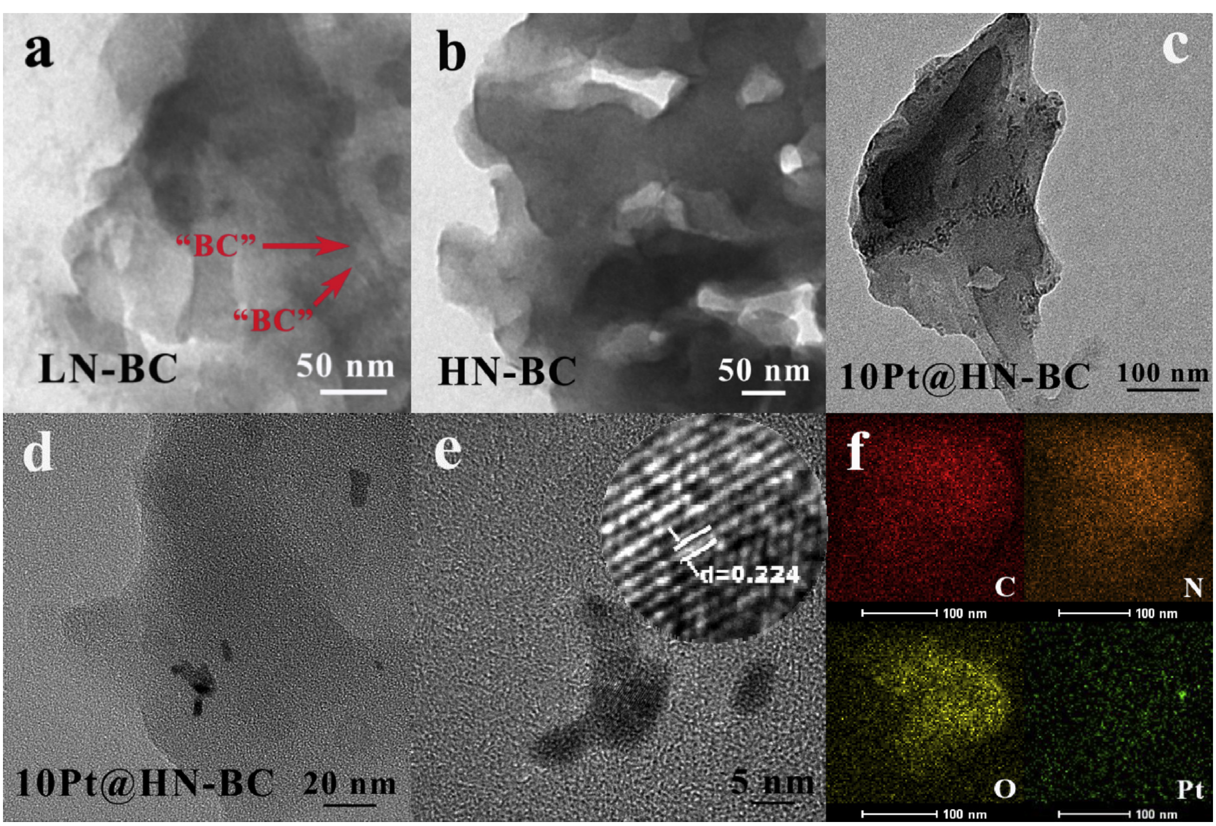

Fig. 3 – TEM images. (a) LN-BC; (b) HN-BC; (c), (d) and (e) 10Pt@HN-BC at different resolutions (low to high); (f) Mapping results of elements.

and c-BC was obtained from calcination of freeze-dried BC at 800 °C for 2 h under nitrogen atmosphere (5 °C min^{-1}).

For preparing the modified glassy carbon electrode (GCE, 3 mm), the 4 mg of as prepared material was dispersed in 1 mL ethanol and water mixed solution (volume ratio of 1:4) with 50 μL Nafion, sonicated until a homogeneous solution was obtained. Then, 5 μL of the above solution was dropped onto the GCE, drying at room temperature.

Before testing, the electrolytes were purged with N_2 for at least 30 min. The synthesis of Pt was via an electrodeposition process in an oxygen free 0.5 M H_2SO_4 with 2 mM chloroplatinic acid hexahydrate for various times in a three-electrode system. The platinum wire and saturated calomel electrode (SCE) were served as the counter and the reference electrodes, respectively. The Pt deposition potential was set at 0.042 V (vs. RHE). The working electrodes were bare GCE, c-BC, PANI, LN-BC and HN-BC modified GCE. The corresponding products were denoted as xPt, xPt@c-BC, xPt@PANI, xPt@LN-BC and xPt@HN-BC (deposition time x = 5, 10, 20, 30, 50 s), respectively. For comparison, 10Pd@HN-BC was obtained via an electrodeposition process in an oxygen free 0.5 M H_2SO_4 with 2 mM palladium chloride for 10 s.

The HER performance of the as-obtained samples was investigated by linear sweep voltammetry (LSV), cyclic voltammetry (CV) and chronopotentiometry measurements performed in the three-electrode system using CHI 660 D electrochemical working station. Electrochemical impedance spectroscopy (EIS) was carried out on an Autolab Nova 1.8 instrument. All the data shown in this work were collected with iR compensation. All potentials were converted to the potentials relative to reversible hydrogen electrode ($E_{RHE} = E_{SCE} + 0.059\ pH + 0.242\ V$). For HER measurements, LSV was conducted within a potential window of 0.1 to −0.4 V with a scan rate of 5 mV s^{-1}. CVs were performed at various scan rates from 20 to 120 mV s^{-1} in 0.5 M H_2SO_4 (potential range: 0.29 to 0.09 V (vs. RHE)). EIS measurement was done at −0.06 V with the frequency of 0.1–10^5 Hz, and the potential amplitude value was 5 mV.

SEM was recorded on an S-4800 field-emission scanning electron microanalyzer (Hitachi, Japan). TEM was carried out on a Tecnai 12 transmission electron microscope (Philips, Netherlands). HRTEM and HAADF-STEM was recorded on a Tecnai G2 F30S-TWIN transmission electron microscope (Philips, Netherlands). XPS measurements were examined with an ESCALAB 250 Xi photoelectron spectrometer using Al Kα radiation (Thermo-Fisher Scientific, US). Raman spectra were acquired using a Micro-Raman spectrometer (Renishaw in Via, U.K).

Results and discussion

Before HER performance evaluation, the samples were fully characterized. In order to understand the morphology

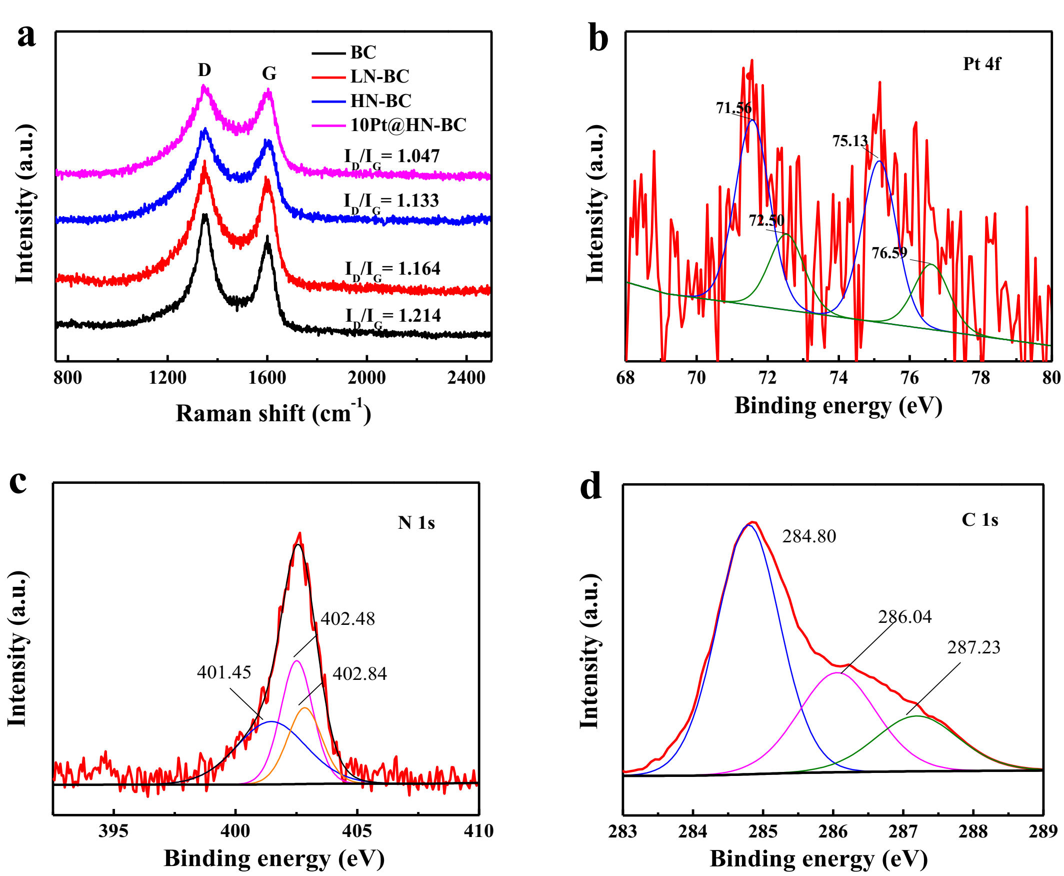

Fig. 4 – (a) Raman spectra of c-BC, LN-BC, HN-BC and 10Pt@HN-BC. XPS spectra; (b), (c) and (d) are Pt 4f, N 1s and C 1s of 10Pt@HN-BC, respectively.

change of the samples, SEM images of c-BC, LN-BC, HN-BC and PANI are shown in Fig. 2. As displayed in Fig. 2a, the c-BC shows a typical nanowire structure, which is similar to the structure of carbon nanotubes. When composited with low-dosage of PANI, nanowire structure decreases because parts of the BC are embedded in the PANI nanosheets (Fig. 2b). As indicated in the inset of Fig. 2c, with increasing the dosage of PANI, BC is fully covered by PANI to form a dark green "BC" (light green for low dosage PANI modified BC). As a result, after pyrolysis treatment under a N_2 atmosphere, PANI and BC melt together. Therefore, nanowire structure cannot be observed in HN-BC as shown in Fig. 2c. For comparison, the image of bare PANI is shown in Fig. 2d, which shows a typical short clubbed structure. The structures of LN-BC and HN-BC are further confirmed by TEM (Fig. 3a and b). The nanowire structure of BC can be observed in LN-BC (Fig. 3a), while no nanowire structure is detected in HN-BC (Fig. 3b). After deposition for 10 s, Pt nanoparticles with diameters ranging from 3.0 nm to 5.0 nm are well dispersed on HN-BC support (Fig. 3c and d) to form the 10Pt@HN-BC nanocomposite. From the high resolution TEM image of 10Pt@HN-BC (Fig. 3e), the detected lattice distance is 0.224 nm, which agrees well with (111) plane of Pt [22]. Mapping results are combined and displayed in Fig. 3f, in which elements of C, N, O and Pt are detected. And the Pt nanoparticles are perfectly dispersed on the support, which is in good agreement to the result shown in Fig. 3c.

The Raman spectra of c-BC, LN-BC, HN-BC and 10Pt@HN-BC are shown in Fig. 4a. Two bands at about 1349 cm^{-1} and 1585 cm^{-1} are D and G band, respectively, which can be attributed to the disorder-induced mode or the defective graphitic structure and E_{2g}-mode from the sp^2 carbon domain, respectively [32]. No obvious Raman shift can be observed in Fig. 4a, however, the change of intensity ratio of the D and G band (I_D/I_G) is evident. The I_D/I_G values of c-BC, LN-BC, HN-BC and 10Pt@HN-BC are 1.214, 1.164, 1.133 and 1.047, respectively. After pyrolysis treatment, the D/G band ratio slightly decreases, which can be attributed to the reduction of disordered carbon or the increase of graphitization [9,11]. The I_D/I_G value decreases further after electrodeposition of Pt on the HN-BC, which may be caused by the decrease of defects in the10Pt@HN-BC.

XPS analysis is also used to evaluate the chemical state of the C, N, O and Pt elements. As displayed in Fig. S1, all the four elements are detected in the full survey of XPS spectrum, which is in accordance to the results of mapping. The XPS analysis also shows the presence of 4.38 wt% Pt on the surface of the 10Pt@HN-BC. Therefore, the absolute Pt loading in 10Pt@HN-BC is 12 μg cm^{-2}. The high-resolution Pt 4f, N 1s and C 1s spectra of 10Pt@HN-BC are shown in Fig. 4b–d. As indicated in Fig. 4b, four main peaks at about 71.56, 72.50, 75.13 and 76.59 eV can be seen, which is assigned to Pt^0 (71.56 and 75.13 eV) and Pt(II) (72.50 and 76.59 eV), respectively [10,33]. The high-resolution N 1s spectrum shown in Fig. 4c reveals

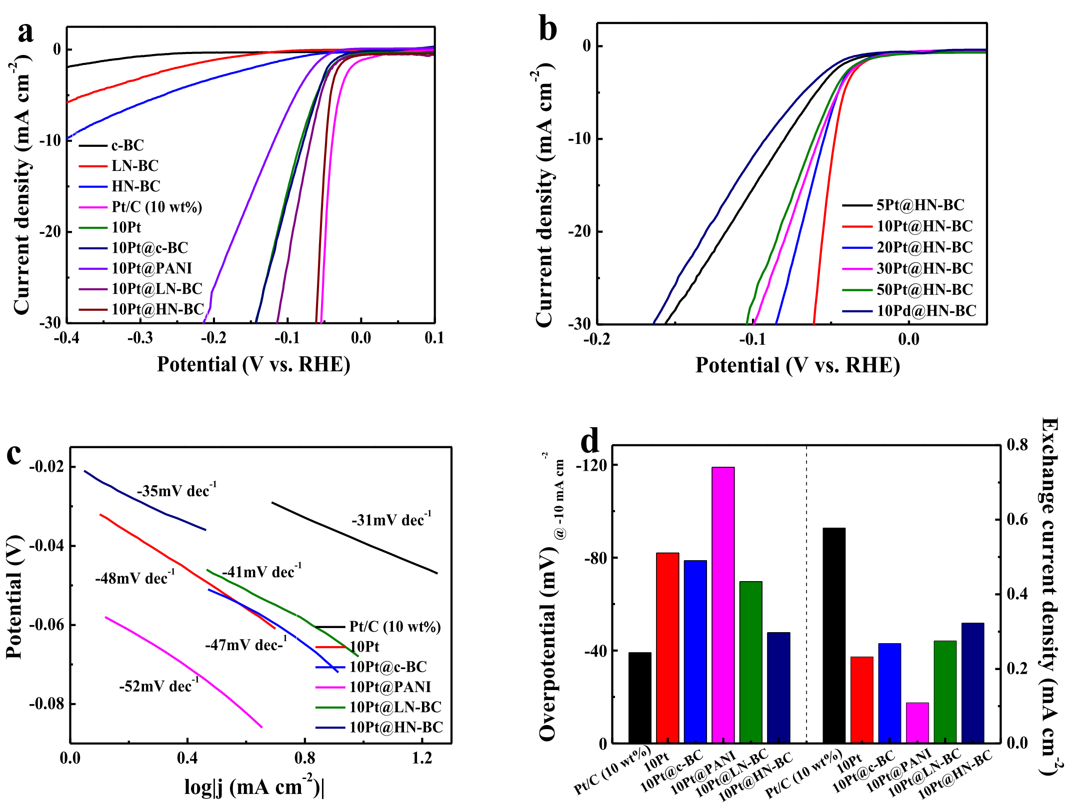

Fig. 5 – (a) Polarization curves of c-BC, LN-BC, HN-BC, 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%) in 0.5 M H_2SO_4; (b) Polarization curves of 10Pd@HN-BC and Pt@HN-BC deposited for various times in 0.5 M H_2SO_4; (c) Tafel plots of 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%); (d) Overpotentials (@-10 mA cm^{-2}) and exchange current densities of 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%).

that pyridinic, pyrrolic and graphitic N are present in 10Pt@HN-BC [10]. From the C 1s spectrum (Fig. 4d), the peaks at 284.80, 286.04 and 287.23 eV correspond to C=C, C=N and C=O, respectively [9].

Linear sweep voltammetry (LSV) is used to investigate the HER activity of c-BC, LN-BC, HN-BC, 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%) in acid (0.5 M H_2SO_4). To obtain the true kinetic activity, iR-corrected data is shown in Fig. S2. It should be noted that all the data used in the following results are collected with iR compensation unless otherwise stated. From the polarization curves shown in Fig. 5a and b, Pt and N dosages are the two main factors affecting the HER activity. Pt free c-BC, LN-BC and HN-BC display poor HER performance, while the Pt-based materials have better electrocatalytic activities (Fig. 5a), which suggests that the HER activity comes mainly from Pt. As far as the Pt@HN-BC is concerned, electrodeposition time results in various Pt dosage on the surface of electrode and will finally affect the HER performance of the Pt@HN-BC. As indicated in Fig. 5b, the HER activity increases first and then decreases obviously when the electrodeposition time is more than 10 s, which suggests that formation of Pt agglomerates when using longer deposition time. Therefore, 10 s is selected as the best deposition time and the corresponding 10Pt@HN-BC with 12 μg cm^{-2} Pt loading is used in the following experiments, unless otherwise stated. For comparison, the HER activity of 10Pd@HN-BC is presented in Fig. 5b, which indicates that the10Pd@HN-BC has poor HER performance. Among all the Pt-based materials obtained in this work (Fig. 5a and b), 10Pt@HN-BC has the best HER activity except for the commercial Pt/C catalyst, which suggests that the increase of N-doping rate causes enhanced HER activity. This is because the higher level of N-doping improves the conductivity of the carbon support. CVs of LN-BC, HN-BC, 10Pt, 10Pt@BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%) in acid (Fig. S3) show the 10Pt@HN-BC (except for Pt/C (10 wt%)) has the largest electrochemical active surface area (ECSA), which is in good accordance to the result of HER activity. The corresponding Tafel slopes of all the Pt-based catalysts (Fig. 5a) in this work are summarized in Fig. 5c. The 10Pt@HN-BC has the lowest Tafel slope (−35 mV dec^{-1}) except for Pt/C (−31 mV dec^{-1}). It is well-known that Volmer, Heyrovsky and Tafel are elementary steps of the HER [34]. When the Volmer, Heyrovsky and Tafel steps control the overall reaction, the theoretical Tafel slopes are −120, −40 and −30 mV dec^{-1}, respectively [34]. Therefore, it can be concluded that H desorption (Tafel or Heyrovsky step) governs the HER kinetics on 10Pt@HN-BC. The results of overpotential (@ −10 mA cm^{-2})

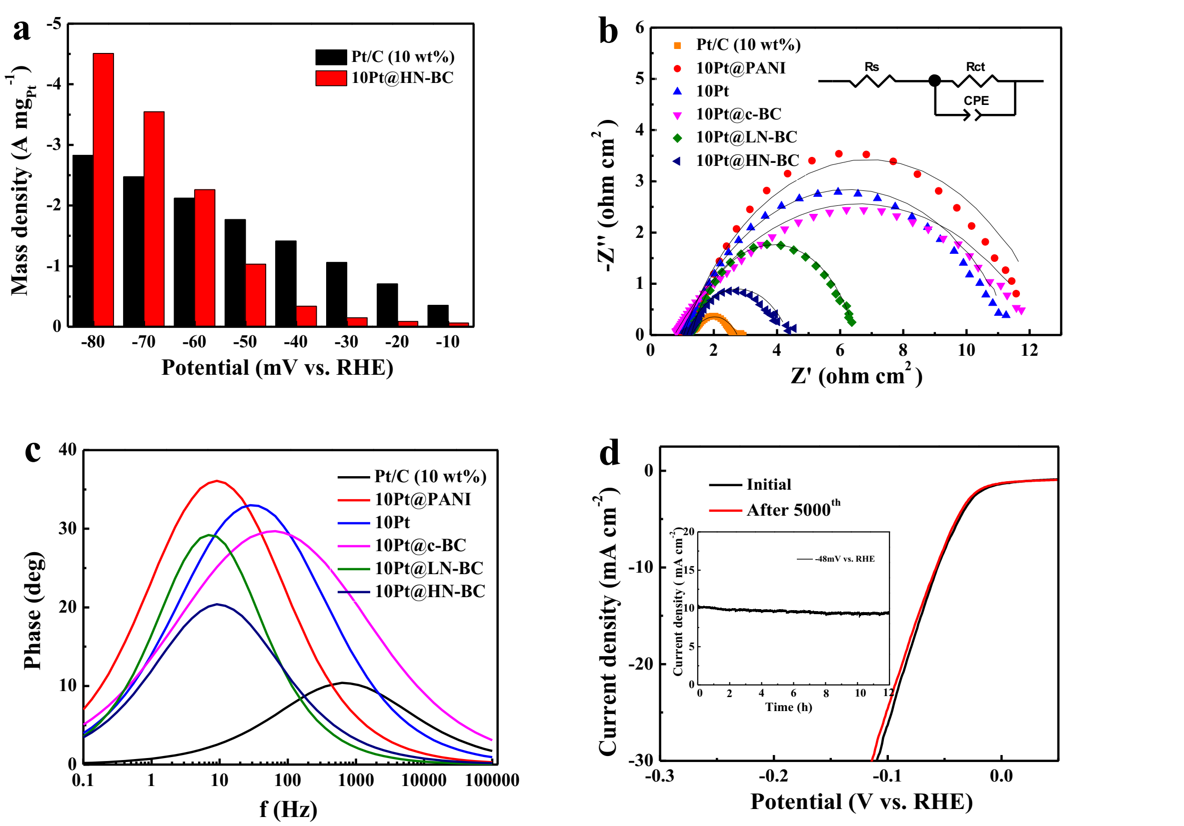

Fig. 6 – (a) Comparison of the mass specific activity of 10Pt@HN-BC and Pt/C (10 wt%) at various potentials (vs. RHE); (b) Nyquist plots of 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%) performed in 0.5 M H_2SO_4, potential: −60 mV (vs. RHE), frequencies: 0.1–10^5 Hz, potential amplitude value: 5 mV; (c) Bode phase plot of 10Pt, 10Pt@PANI, 10Pt@c-BC, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%), respectively; (d) Polarization curves of 10Pt@HN-BC initially and after 5000 cycles under 0 to −0.1 V (vs. RHE) at 5 mV s^{-1} in 0.5 M H_2SO_4. Inset: Time-dependent current density curve of 10Pt@HN-BC in 0.5 M H_2SO_4 under a static overpotential of −48 mV (vs. RHE).

and exchange current density are calculated and shown in Fig. 5d. For Pt-based materials obtained in this work, 10Pt@PANI (-118 mV dec^{-1}) and 10Pt@HN-BC (-35 mV dec^{-1}) have the largest and lowest overpotentials in acid, respectively. Moreover, the 10Pt@HN-BC (except for Pt/C (10 wt%)) has the largest exchange current density of 0.323 mA cm^{-2}, which is higher than that of 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC. Although the HER performance of 10Pt@HN-BC is slightly lower than that of Pt/C (10 wt%), the dosage of Pt in 10Pt@HN-BC is obviously lower than that in Pt/C (10 wt%).

To further evaluate the HER performance, comparison of the mass specific activity of 10Pt@HN-BC and Pt/C (10 wt%) at various potentials (vs. RHE) is shown in Fig. 6a. As displayed in Fig. 6a, at overpotentials less negative than -50 mV, the mass specific activity of 10Pt@HN-BC is lower than that of Pt/C (10 wt%). However, when the overpotential is increased to -60 mV, the 10Pt@HN-BC has mass specific activity of -2.25 A mgPt^{-1}, which is higher than that of Pt/C (10 wt%). The HER performances of 10Pt@HN-BC and other reported Pt-based or Pt-like electrocatalysts in acid are compared and summarized in Table 1 [9,35–43]. From Table 1, the 10Pt@HN-BC obtained in this work has quite comparable HER performance than that of the Pt-based or Pt-like HER catalysts reported recently.

Electrochemical impedance spectroscopy (EIS) is conducted to further confirm the kinetics process [44]. The classic Nyquist plots (scatter diagram) are shown in Fig. 6b. As displayed in Fig. 6b, the experimental and fitted data are of great relevance, while small inconsistency between the experimental and fitted results can be observed at low frequencies, which may be due to the complex mass transformation. Table 2 presents the values of the simulated electrochemical parameters using the equivalent circuit shown in Fig. 6b (inset), where R_s is the solution resistance, R_{ct} is the resistance of the charge-transfer step, CPE_T is the capacitance parameter and n is parameter associated with deviation from the ideal capacitance, which value may between 0 and 1. The Nyquist diagram looks like slightly capacitive semicircle. Clearly, the simulated electrochemical results confirm that a remarkable decrease in charge transfer resistance (R_{ct}) from 10.26 Ω cm^2 (10Pt@PANI) to 4.98 Ω cm^2 (10Pt@LN-BC) and then to 2.61 Ω cm^2 (10Pt@HN-BC), indicating the fastest electro-catalytic reaction rate for HER on the 10Pt@HN-BC electrode in acid solution [45]. Only one peak at low frequencies can be observed (except Pt/C) from the Bode phase plot (Fig. 6c), which is related to the kinetics of the electrochemical hydrogen evolution reaction. The Bode plot also confirms typical one-time-constant behavior related to smooth surfaces with one dominant reaction.

The 10Pt@HN-BC also shows good stability in acid. As shown in polarization curve (Fig. 6d), slight cathodic current loss can be observed after 5000 cycles under 0 to -0.1 V (vs. RHE) at 5 mV s^{-1} in 0.5 M H_2SO_4. From the *j*-*t* curve (Inset of Fig. 6d), only 6.5% degradation is detected after electrolysis in 0.5 M H_2SO_4 under -48 mV for 12 h.

As the electrochemical double layer capacitance (C_{dl}) is a crucial parameter to investigate the electrocatalytic performance of HER catalyst, the C_{dl} of 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%) is also calculated based on the CV curves shown in Fig. 7a and S4. The corresponding capacitive current at 0.195 V (Fig. 7a and S4) as a

Table 1 – HER performances of 10Pt@HN-BC and other reported Pt-based or Pt-like electrocatalysts in acid.

Material	Electrolytes	Relative Pt or Pd loading (wt%)	Absolute Pt or Pd loading (μg cm^{-2})	Onset potential (mV) @ -1 mA cm^{-2}	Overpotential (mV) @ -10 mA cm^{-2}	Tafel slope (mV dec^{-1})	Exchange current density (mA cm^{-2})	C_{dl} (mF cm^{-2})	TOF (H_2 s^{-1})	Reference
10Pt@HN-BC	0.5 M H_2SO_4	4.38	12	−18	−47	−35	0.323	11.5	1.56@-50mV	This work
PtCoFe@CN	0.5 M H_2SO_4	4.6	1.3	N/A	−45	−32	N/A	10.5	N/A	[9]
Pt/BCF	0.5 M H_2SO_4	0.87	N/A	N/A	−55	−32	N/A	N/A	N/A	[35]
Pt NIs@f-MWCNT	0.5 M H_2SO_4	7	3.8	−33	−84	−21.2	N/A	76×10^{-3}	6.3@-70mV	[36]
Pt-Ag/SiNW	0.5 M H_2SO_4	4.1	7.0	N/A	−135	−70	0.165	9.99×10^{-3}	6.3@-200mV	[37]
polyTT-Pt	0.5 M H_2SO_4	2.91	8.6	−8	−67	−37	0.28	N/A	N/A	[38]
CNT/DenPtNPs	0.5 M H_2SO_4	1.03	N/A	−16	−50@-2.5 mA cm^{-2}	−42	N/A	N/A	N/A	[39]
3%Pt/WS_2	0.5 M H_2SO_4	3	N/A	N/A	−80	−55	N/A	N/A	N/A	[40]
$Pd_{71}Cu_{29}$/C	0.1 M H_2SO_4	N/A	N/A	N/A	−75	−48	0.38	N/A	N/A	[41]
Pt–TiO_2–N-rGO	0.1 M H_2SO_4	6.33	N/A	N/A	−300@-126 mA cm^{-2}	−32	0.22	N/A	N/A	[42]
Pt-CNSs/GO	0.5 M H_2SO_4	47.25	80.3	−18	N/A	−29	0.18	N/A	N/A	[43]

Table 2 – EIS parameters of 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%).

	R_s (Ω cm^2)	R_{ct} (Ω cm^2)	CPE_T (S s^n cm^{-2})	n	$C_{dl\ (EIS)}$ (mF cm^{-2})	$C_{dl\ (CV)}$ (mF cm^{-2})
Pt/C (10 wt%)	1.37	1.14	0.030	0.80	11.2	11.8
10Pt	1.32	9.01	0.013	0.80	4.56	5.60
10Pt@BC	1.33	9.73	0.014	0.76	4.02	4.90
10Pt@PANI	1.33	10.26	0.011	0.79	3.28	3.50
10Pt@LN-BC	1.37	4.98	0.021	0.82	8.74	8.10
10Pt@HN-BC	1.33	2.61	0.029	0.79	10.6	11.5

*EIS measurements were performed at a potential of −0.06 V vs. RHE.

function of scan rate for 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%) in 0.5 M H_2SO_4 is displayed in Fig. 7b, in which straight lines are obtained. The calculated C_{dl} of 10Pt@HN-BC is 11.5 mF cm^{-2}, which is slightly lower than that of the Pt/C (10 wt%) (11.8 mF cm^{-2}) and much higher than that of other Pt-based catalysts obtained in this work. For comparison, $C_{dl\ (EIS)}$ was calculated using the following expression: $C_{dl} = [CPE_T/(R_s^{-1}+R_{ct}^{-1})^{1-n}]^{1/n}$ [46]. The $C_{dl\ (EIS)}$ and $C_{dl\ (CV)}$ are summarized in Table 2, in which the $C_{dl\ (CV)}$ agrees well with the $C_{dl\ (EIS)}$.

Turnover frequency (TOF) is another crucial parameter to evaluate the performance of catalyst. According to the literatures [3,47,48], the under potential deposition (UPD) of copper on Pt and Ru is an ideal way to calculate their corresponding

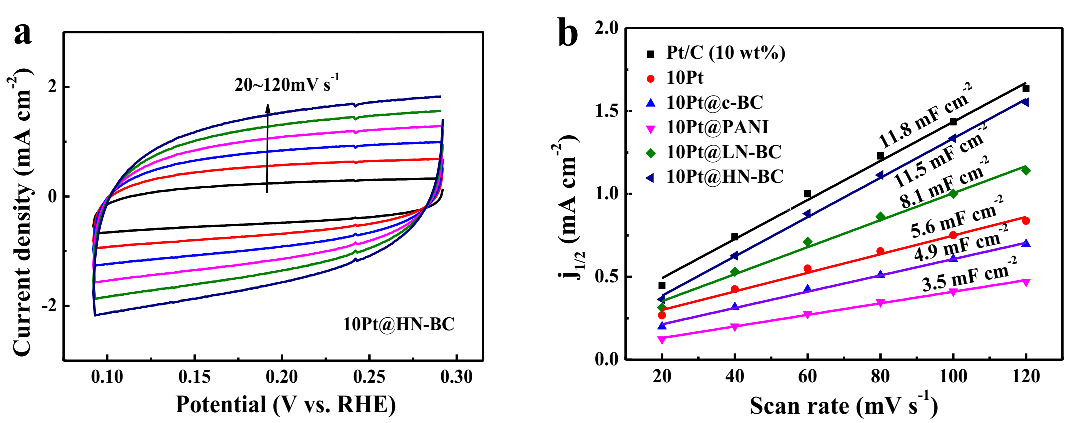

Fig. 7 – (a) CVs of 10Pt@HN-BC with various scan rates from 20 to 120 mV s^{-1} in 0.5 M H_2SO_4, potential ranges: 0.29–0.09 V (vs. RHE) (b) The capacitive current at 0.195 V (vs. RHE); as a function of scan rate for 10Pt, 10Pt@c-BC, 10Pt@PANI, 10Pt@LN-BC, 10Pt@HN-BC and Pt/C (10 wt%) in 0.5 M H_2SO_4.

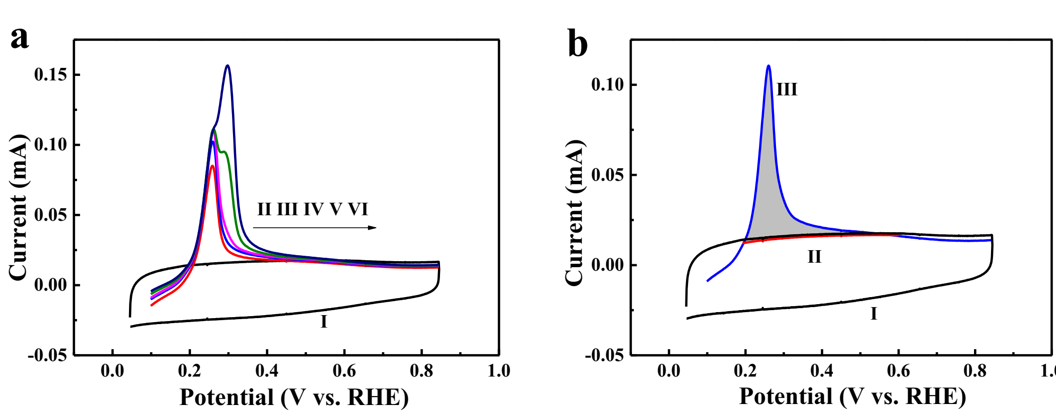

Fig. 8 – (a) Copper UPD in 0.5 M H_2SO_4 in the (I) absence and (II–VI) presence of 2 mM $CuSO_4$ on 10Pt@HN-BC. Curve I refers to CV and curves II–VI to anodic linear sweep voltammetry (ALSV). For curves II–VI, the electrode was polarized at 0.225, 0.220, 0.215, 0.210 and 0.205 V for 100 s to form the UPD layers, respectively; (b) Copper UPD in 0.5 M H_2SO_4 in the (I, II) absence and (III) presence of 2 mM $CuSO_4$ on 10Pt@HN-BC. Curves II and III refer to ALSV and curve I to CV. For II and III, the electrode was polarized at 0.215 V for 100 s to form the UPD layer. Sweep rate: 20 mV s^{-1}. Potential ranges: 0.05–0.85 V (vs. RHE) for CVs, 0.10–0.85 V (vs. RHE) for ALSV.

active sites [3]. If copper stripping charge (Q_{Cu}) was obtained, the value of n could be calculated from the Equation ($n = Q_{Cu}/2F$, n is the number of active sites (mol), F is the Faraday constant). Finally, the TOF can be obtained from Equation ($TOF = I/(2Fn)$, I is the current (A) during linear sweep measurement). To obtain the TOF, the 10Pt@HN-BC is kept in a 0.5 M H_2SO_4 solution with and without 2 mM $CuSO_4$ at different potentials for 100 s to examine the influence of the polarization potential on the Cu stripping voltammetric charge (Fig. 8a), which suggests that the highest charge is obtained at 0.215 V. According to the result shown in Fig. 8b, the estimated number of active sites of 10Pt@HN-BC is 4.25×10^{-8} mol cm^{-2}. Therefore, the TOF values of 10Pt@HN-BC @ −25 and −50 mV are 0.17 and 1.56 H_2 s^{-1}, respectively.

Conclusion

In summary, a platinum nanoparticles modified nitrogen-doped carbon nanofiber derived from BC was obtained via electrochemical deposition. The LN-BC and HN-BC works as support. Platinum nanoparticles with diameters ranging from 3 to 5 nm can be well dispersed in the support. After pyrolysis and Pt deposition, the graphitization is increased. The HER performance of xPt@LN-BC and xPt@HN-BC is fully investigated. The 10Pt@HN-BC with 4.38 wt% Pt loading shows the best HER catalyst among the xPt@HN-BC in acid. The onset potential (@ −1 mA cm^{-2}) and overpotential (@ −10 mA cm^{-2}) of the 10Pt@HN-BC in 0.5 M H_2SO_4 is −18 and −47 mV, respectively. The Tafel slope and double layer capacitance is −35 mV dec^{-1} and 11.5 mF cm^{-2}, respectively. The exchange current density of 10Pt@HN-BC is 0.323 mA cm^{-2}. The TOF values of 10Pt@HN-BC @ −25 and −50 mV are 0.17 and 1.56 H_2 s^{-1}, respectively. The low dosage of Pt and well durability in acid make it a promising candidate for HER catalyst.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.ijhydene.2018.02.054.

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